Words in Context
Explaining Foreign Expressions with Sample Sentences

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Abstract

In recent years we are more then ever confronted with new words. They are used to describe new things and concepts. More and more people are learning foreign languages and foreign words can be found in our everyday life. Proficient language users can usually infer the meaning of newly encountered word from context in which it is presented. Sometimes however, the context is not sufficient or misleading, causing confusion and forcing the user to search for explanation. Dictionaries have proven themselves a great help, especially by providing sample sentences presenting the most appropriate use of a given word. However, they have limited capacity and can't adapt to specific needs. In this paper we present several problems with existing dictionaries and explain why they often fail to achieve our goals. Finally, we propose a solution to these problems by presenting the idea of a system that would provide learners with sample sentences that meet their needs.

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1 Introduction

With the ongoing globalisation we are daily confronted with more and more foreign words. This affects all people, who are more often than ever confronted with loanwords in their daily lives. Kister claims[1] that English language may grow by as much as 25000 words/year. More recent analysis found the growth rate at about 8500 words/year for the last 50 years[2]. And even if most of them are only interesting for relatively small groups of people, some of those words will find their way into the mainstream language.

The problem of foreign words is even more difficult to language learners, who are expected to quickly memorize large amounts of them. But textbooks follow their own path, not always matching the needs of students. Regardless of their background students

are forced to follow standard courses while being more in need of specific vocabulary specific to their field of interests.

Typically people use one of several typical approaches when confronted with a new word. Those proficient in the language can usually guess the meaning from context [3]. In fact, there are claims that most of vocabulary may be acquired this way[4]. After a person encounters a word several times it is usually absorbed into vocabulary[5].

Language students may not have sufficient vocabulary to guess the meaning from context. According to Laufer[6] best results are achieved when over 95% words are known to the reader. Students with lower proficiency will have to refer to dictionary. However, bilingual dictionaries introduce other problems, like narrowing word meanings or inhibiting definition skills[7]. Using monolingual dictionary may be a solution, but this again requires large vocabulary to start with.

More direct approach to learning new words involve flashcard study, often augmented by spaced repetition as proposed by Leitner[8]. Availability of computers allowed individual tracking and scheduling of each flashcard further increasing effectiveness of this method as shown by Wozniak[9].

There are many approaches to flashcard creation. Some examples are:

- word based or decontextualized - intended to teach specific words without context
- sentence based or contextualized - intended to use sentence context to teach one or more words
- recognition or recall based - depending on whether the learned material constitutes question or answer
- using multimedia aids - audio, images or even video

Which approach is best depends on many factors, and is often difficult to predict. Especially whether sentence or word based approach is better seems to be often source of heated debate without clear answer[4, 5, 10].

Sentence based approach, usually relying only on recognition, is similar to examples in monolingual dictionaries. It uses sentence context to rely meaning of the words. Still, selecting sentence requires picking enough words familiar to student, so to provide basis for guessing the unknown ones.

We present an idea for creating sample sentences specifically tailored for needs of individual users.

2 Possibilities of digital age

In past most of language studies were bound to paper. Books and dictionaries were printed, students wrote down the vocabulary lists and practiced using pen and paper. Access to information was limited by amount and availability of paper copies of desired materials. Today, with availability of computers and information networks this has changed significantly. Digital dictionaries are small, but can contain more information or multimedia materials which was not possible in their paper counterparts. Students can use computers as study help to create easier to manage word lists or utilize flashcard software capable of utilizing sophisticated scheduling algorithms[9]. Internet provides access to uncounted sources of information in various forms, like Classical literature is available through projects like Project Gutenberg[11] or Aozora Bunko[12], ebooks, news sites, blogs and many other. This creates new opportunities in vari-

ous fields.

3 Previous research in this area

Most research on sentence generation concentrated on different goals than those addressed in this paper. Popular topic among researchers are chatterbots, starting with ELIZA in 1966[13]. Still, chatterbots concentrate on providing natural conversation or answering users questions. Dowding et al.[14] researched the topic of targeted help in a spoken dialogue system, where example sentence is used as feedback to users utterance. But the goal was to provide feedback that can be understood by user, not to encourage learn new terms.

A research to generate example sentences was undertaken by Liu et al.[15], who used Google N-gram corpus to create example contexts for vocabulary training for children. Some problems were identified, but overall initial results were optimistic.

4 Research concept

This paper presents an idea of generation of example sentences using several data sources. Current vocabulary - or what the student already knows and understands. Target vocabulary - what the student wants to know after practicing with the example sentences. And additional data set, which is used to aid the creation of the example sentences.

4.1 Determining current vocabulary

To create personalized sample sentences, tailored to particular student, it is first necessary to determine the current vocabulary. This will constitute a baseline of students current knowledge. While student may be able to provide list of words he is familiar with, other solutions have to be also considered, like using more general sources. Possible sources may include:

- Software flashcards. Depending on software used it may be possible to just extract list of words up to the full learning history. The later option has the additional benefit of identifying words easier and more difficult for particular student.
- Lists of vocabulary used in traditional textbooks. It won't take into account differences between words, but may provide a good basis.
- Sample materials student considers easy to read. It can be expected that he is familiar with vocabulary used in those materials.

Depending on what data can be analyzed different amount of information can be extracted. Frequency analysis of word occurrences, typical sentence patterns or grammar forms can indicate students skill level.

4.2 Identifying target vocabulary

The next task is to identify target vocabulary. This can be approached similar as determining current vocabulary using readily available sources:

- Lists of words
- Sample texts the student expects to master

Interestingly, this is very similar to determining current vocabulary, and it may be possible to perform similar analysis on them. The main difference would be the use of extracted data - while student should be familiar with data from current vocabulary, target

vocabulary data should only be introduced carefully and gradually. Important to note is that is special case this may constitute just one word.

4.3 Selecting additional data

The language we use depends on the demands of situation[16]. So the sample sentences should be tailored to the needs of particular student. To meet this goal it is again helpful to make distinction between several possible sources of data. It is expected that in most cases three groups of sources will be necessary.

1. First group, consisting of data with the target vocabulary as described in section 4.2. While it is possible that this source will be of significant size, it is also possible that a short text, or even a list of words are used. In such case not enough data would be available to select sufficient number (if any) sample sentences. If possible, this data should be also used to extract sentences for use as existing sentences.

2. Corpus for extracting existing sentences. It is preferred that it is also tailored for student's needs. As students usually intend to study specific vocabulary, they may find sample sentences extracted from texts covering related topics more valuable. They are likely to use similar vocabulary or even specific grammatical expressions. Using a more generic corpus may result in sentences where a target word would appear in otherwise unrelated sentence.

3. Source for data for modifying sample sentences selected from corpus described above. This data may be extracted from any source, like a larger collection of natural language text, or completely different source like lexical database.

It is possible that in specific cases some of these sources can overlap. It remains to an area for further research to determine how such combinations would affect resulting sentences.

4.4 Creating sentences

For each word in target vocabulary (not found in current vocabulary) at least one, but preferably more, candidate sentences have to be generated using data from existing sentences corpus. Each sentence should be verified to use only vocabulary from current vocabulary list, except for small number of new words. Provided sufficiently large corpus is available it may be possible to find sentences that can be used unmodified, directly satisfying the constraints.

If the constraints are not fulfilled the sentence has to be discarded or modified – some words may need to be replaced to match the criteria, while care must be taken to preserve the logical and grammatical consistency of the sentence. Additional data described in section 4.3 can be used in this step. Finally, the modified sentence should be verified for it's usability as example sentence. Liu et al. [15] suggested several constraints, like being grammatically correct and complete (not depending on additional context), varied or not redundant (avoiding very similar sentences).

Depending on target environment where the sentences will be used some additional considerations may arouse. If sentences will be used predictable order, then it will be possible to re-use words introduced by earlier sentences as help for understanding later ones. It the sentences are intended for computer assisted vocabulary learning it may be possible to use data from such system as feedback to try to identify problematic sentence patterns or to further improve

the generated sentences by adding additional examples for problematic words.

5 Additional considerations

5.1 Identifying suitable sentences

The system described above requires a method to identify suitable sample sentences. This may be a major challenge to accomplish a working system. Let's look closer at some of the problems.

In real word sentences are usually part of some larger works. Often they can only convey meaning by appearing in larger context. Sentences in natural language vary between short statements like "*And that was all.*" to very complex sentences. Example sentences are usually expected to be as short as possible, while still preserving sufficient information to be self-explanatory.

Let's consider following sentence: "*They were all about him and the fire, in a narrow circle, and he could see them plainly in the firelight lying down, sitting up, crawling forward on their bellies, or slinking back and forth.*" It seems too complex to be used as a simple example, while still not appearing sufficiently self-explanatory.

Another sentence "*Henry sat down on the sled and watched.*" avoids above issues presenting a short self-contained statement. It seems much more likely that it can be used for purpose of example sentence, but still does not provide any hints to explain used words.

5.2 Missing words

The issues mentioned previously may make it difficult to identify appropriate existing sentence that could be directly used to demonstrate particular word even using a large corpus. This problem can be mitigated by introducing modifications to existing sentences. Possible approaches include extracting simple parts from complex sentences or substituting some words. But such modifications are risky, as it's easy to break logical consistency of a sentence.

Let's consider a modification of a sentence from section 5.1: "*The wolves were all about Henry and the fire.*" After leaving only the first clause and clarifying subject and object this sentence seems much better suited for simple example. It should be also considered, that while this modification was relatively simple, using only a small additional context to determine the subject and object, some changes are much more complex.

Let's consider again sentence "*I like cars painted red.*". Assuming the student knows all words used in this sentence except "*red*" he will be able to infer that "*red*" is color. A possible modification of this sentence could change the color, resulting in "*I like cars painted green.*". While this would also show that "*green*" is a color student would still not know *which* color it is. Adding sentence: "*Grass is green.*" where student already knows word "*grass*", he will be able to correlate the word with the correct meaning.

However modifying the second sentence is similar manner as the first one it is possible to end with the misleading "*Grass is red.*". A potential solution to avoid such problems would be to use commonsense database like ConceptNet[17] to identify correct word correlations.

6 Summary

Natural language will always surprise us with new uses for old words. While dictionaries try to pro-

vide definitions for words, they often fail to convey specific nuances or atypical uses. Native speakers often notice students using formally correct sentences that still sound strange to them - nobody would say it this way. Using sample sentences shows words in their natural environment, ready to use in a specific scenario. And while they certainly can't solve all problems associated with vocabulary learning, they certainly can be a big help. Making the sample sentences more tailored to individual needs may further increase their usefulness.

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